

Taro UH

Genotyping by Sequencing (GBS) UNEAK Pipeline

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For questions regarding your project/data specifically please contact gbsdata@cornell.edu and
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For general questions regarding the GBS pipeline, GBS data analysis and TASSEL, please use the TASSEL
google group here, so that others may benefit from your questions and answers:

<https://groups.google.com/forum/?fromgroups#!forum/tassel>

additional GBS resources are provided below

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Sample Details

189 uniquely named samples (190 total) and 2 blank(s) digested with enzyme(s) PstI

Library Plate	Library Plate ID	Filename (flowcell_lane_fastq.gz)	md5sum
1_1025_Cross	450018209	C3008ACXX_8_fastq.gz	e0062287dc9a04dc45032175b8e88413
2_Hawaiian	450018210	C2YWBACXX_6_fastq.gz	e694672c76096c208d39b7d8d906f3b6

Analysis Notes

The GBS UNEAK analysis pipeline (Tassel Version: 3.0.160 Date: August 22, 2013) was run on these samples

If appropriate, VCF snps were merged with Tassel Version: 4.3.0 (August 22,2013) – see description of VCF files.

An overview of GBS and the GBS pipeline can be found here: <http://www.maizegenetics.net/Table/Genotyping-By-Sequencing/>. See the following sections for a list of options used for this analysis.

We are providing:

- 1) SNP calls output from the GBS Bioinformatics pipeline (see description of data files below).
- 2) TOPM (tags-on-physical-map) file (see description of data files below).
- 3) Keyfile(s) used to associate barcodes with sample IDs while running the GBS pipeline.
- 4) This report summarizing the SNP calling procedure, including software versions used
- 5) XML configuration files that can be used (with the software versions indicated) to reproduce the results summarized here. XML files can be used by invoking the corresponding TASSEL plugin in the following manner (adjusting the maximum amount of memory -Xmx if necessary)

```
run_pipeline.pl -Xmx8G -configFile xmlfile.xml
```

as directory structures and file locations differ, please edit the corresponding options appropriately if you wish to utilize the .xml file for your own analysis.

***Please note**, the GBS pipeline is continually being developed, and as such the results from this run can only be reproduced with the GBS pipeline version noted above, along with the options listed in the following section or utilizing the xml files distributed with your data.

Description of Data Files

We are providing SNP calls in both HapMap and VCF format. Currently SNP calling as implemented for hapmap and VCF files are independent, and as such variations between the two files can be expected.

HapMap files

HapMap is the standard file format generated by the TASSEL GBS pipeline. We are providing two sets of HapMap files:

1. A set without post SNP-calling filtering (located in the folder “hapmap/unfiltered”)
2. A set with additional filtering on missingness and allele frequency (located in the folder “hapmap/filtered”).

Within both folders you will find either one or two sets of files, depending on whether your project included identically named samples.

1. A hapmap file with separate SNP calls for identically named taxa (c+.snpmmerged.hmp.txt)
2. A hapmap file with SNPs merged for identically named taxa. (c+.taxaMerged.hmp.txt)

The naming convention of these HapMap files is c+, where + is the chromosome number (where available). In the case of an assembly that is not in the chromosome/pseudomolecule stage there will be one or more files, but “+” will not correspond to a chromosome number. Columns 3 and 4 in the HapMap file correspond to chromosome (or contig/scaffold) and position, respectively.

For GBS libraries sequenced multiple times, identical samples are merged prior to SNP calling. Identically named samples that did not originate from the same library are merged after SNP calling.

Here we have filtered on missingness only (see the options listed below for the plugin GBSHapMapFiltersPlugin). We highly recommend filtering your data with parameters that are appropriate for your species/project. HapMap files can be easily opened and filtered in TASSEL 4.0.

http://www.maizegenetics.net/index.php?option=com_content&task=view&id=89&Itemid=119

The header line of the hapmap file contains information on the information contained within.

VCF files

VCF is an alternative format for holding SNP information that retains information on depth of coverage for each allele, and can be output from the GBS pipeline. SNPs are called using the algorithm from The format is not currently supported for analysis with TASSEL 3.0, although support for VCF in TASSEL 4.0 has now been implemented.

We are providing one file (all.mergedSNPs.vcf.gz) that contains all SNP calls, and has been compressed with bgzip. You can unzip this file with the command gunzip on Linux/Unix, or with any of several software options on windows (7-zip is a good one).

The header lines of this file (beginning with #) contain information regarding the contents of the file. Please note that the alleles are in the format major/minor rather than reference/alternate.

Multiply sequenced or identically named samples are not merged during VCF SNP calling, but identically named samples are merged after SNP calling with the experimental **TASSEL 4.0 plugin "MergIdenticalTaxaPlugin"** as a courtesy. The xml file for this plugin is included in the data distribution.

You can also work with VCF files output from the GBS pipeline using VCFtools (<http://vcftools.sourceforge.net/>) and other software platforms. VCFtools is currently only available for Linux/Unix platforms. If you do not have access to a Linux computer and wish to use VCFtools, you can purchase machine hours on CBSU's BioHPC Computing Lab (<http://cbsu.tc.cornell.edu/lab/Pricing.aspx>).

For heterozygous materials and diversity studies (in cases where there is little information at the population level to leverage for filtering SNPs), and especially where sequence coverage is low (as is typical for GBS data), we strongly suggest working with the VCF files and utilizing the allelic read depth and/or SNP quality information for filtering.

TOPM file:

This file contains the actual tag sequences and tag pairs. It is a TOPM "tags on physical map" file. Because there is no physical map, this file uses arbitrary positions along a fake chromosome to anchor tag pairs. You will find

In the header line, listed by column:

- 1) number of tags in the file,
- 2) number of "longs" used to store a tag. 1 long = 32 bp. 2 longs = 64 bp.
- 3) Maximum number of variants stored per tag (meaningless for this SNP discovery pipeline)

In each subsequent line, listed by column:

- 1) The tag sequence
- 2) The length of the tag (tags shorter than 64 bp are padded with "A" in this file, but the extra padding is not used during SNP calling)
- 3) The number of genomic positions the tag aligns to (99 means multiple >1 positions).
- 4) Chromosome (* means no unique alignment position)
- 5) Strand (1 or -1, * means no unique alignment position) [note: for tags that align to a unique position but on the minus strand End Position < Start Position]
- 6) Start Position of the barcoded end (* means no unique alignment position)
- 7) End Position of the common adapter end, (* means no unique alignment position)
- 8) Divergence (edit distance from the aligned position on the reference genome, in this case there is no reference pipeline and the edit distance is always 1, as tag pairs are tags that differ by 1bp).
- 9) The remainder of each row consists of position offsets and the alleles for the 8 variants. They are meaningless here, and will all be "**"

There are two tags for each start/stop position, these represent tag pairs, which are described in the original publication (Lu et al 2013).

You can use the SNP chromosome and position from the hapmap (columns 3 and 4) or VCF file (columns 1 and 2) in conjunction with the TOPM file to determine which tag pair from which it originated.

GBS reference pipeline options used for analysis

(not including file direction options – please see xml files for full list of options used)

plugin	option	value	description
FastqToTagCountPlugin	c	1	Minimum number of times a tag must be present to be output. Default: 1
FastqToTagCountPlugin	s	300000000	Max good reads per lane. (Optional. Default is 300000000).
MergeMultipleTagCountPlugin	c	5	Minimum number of times a tag must be present to be output. Default: 1
TagCountToFastqPlugin	c	1	Minimum count of reads for a tag to be output (default: 1)
UtagCountToTagPairPlugin	e	0.03	Error tolerance rate in the network filter. (Default: 0.03)
UExportTagPairPlugin	d	1000	(or --distance) distance to pad tag pairs by (default: 1000)
FastqToTBTPlugin	y	-y	output to tagsByTaxaByte (tag counts per taxon from 0 to 127) instead of tagsByTaxaBit (0 or 1)
FastqToTBTPlugin	c	1	Minimum taxa count within a qseq file for a tag to be output. Default: 1
MergeTagsByTaxaFilesPlugin	s	200000000	Maximum number of tags the TBT can hold while merging (default: 200000000)
TagsToSNPByAlignmentPlugin	y	-y	Use byte-formatted TBT file (*.tbt.byte)
TagsToSNPByAlignmentPlugin	errRate	0.01	Average sequencing error rate per base (used to decide between heterozygous and homozygous calls) (default: 0.01)
TagsToSNPByAlignmentPlugin	mnLCov	0.1	Minimum locus coverage i.e. the proportion of taxa with at least one tag at the locus. Default: 0.1
TagsToSNPByAlignmentPlugin	mxSites	2000000	The maximum number of SNPs per chromosome for hapmap files (default = 2000000)
TagsToSNPByAlignmentPlugin	mnMAC	999	Minimum minor allele count. Defaults to 10. SNPs that pass either the specified minimum minor allele count (mnMAC) or frequency (mnMAF) will be output.
TagsToSNPByAlignmentPlugin	mnMAF	0.01	Minimum minor allele frequency. Defaults to 0.01. SNPs that pass either the specified minimum minor allele frequency (mnMAF) or count (mnMAC) will be output.

plugin	option	value	description
MergeDuplicateSNPsPlugin	misMat	0.05	Threshold mismatch rate above which the duplicate SNPs won't be merged. Default: 0.05.
MergeDuplicateSNPsPlugin	callHets	-callHets	When two genotypes at a replicate SNP disagree for a taxon call it a heterozygote. Defaults to false (=set to missing)
MergeIdenticalTaxaPlugin	hetFreq	0.8	cutoff frequency between het vs. homozygote calls (default = 0.8)
FastqToTBTPlugin	y	-y	output to tagsByTaxaByte (tag counts per taxon from 0 to 127) instead of tagsByTaxaBit (0 or 1)
FastqToTBTPlugin	c	1	Minimum taxa count within a qseq file for a tag to be output. Default: 1
MergeTagsByTaxaFilesPlugin	s	200000000	Maximum number of tags the TBT can hold while merging (default: 200000000)
tbt2vcfPlugin	ak	3	Maximum number of alleles that are kept for each marker across the population default: 3
tbt2vcfPlugin	mnLCov	0.0	Minimum locus coverage (proportion of Taxa with a genotype) (default: 0.0)
tbt2vcfPlugin	mnMAF	0.0	Minimum minor allele frequency (default: 0.0)
MergeDuplicateSNP_vcf_Plugin	ak	3	Maximum number of alleles that are kept for each marker across the population default: 3
GBSHapMapFiltersPlugin	mnSCov	0.8	Minimum site coverage (default: no filter)
GBSHapMapFiltersPlugin	mxMAF	1	Maximum minor allele frequency (default: 1.0 = no filter)
GBSHapMapFiltersPlugin	mnTCov	0.1	Minimum taxa coverage (default: no filter)
GBSHapMapFiltersPlugin	mnMAF	0.01	Minimum minor allele frequency (default: 0.0 = no filter)

Reads and Tags

Summary of Reads and Tags found in each Sequencing Lane

**Please note that blanks must be named "blank," case insensitive to be automatically excluded from summary calculations, and to be excluded from the failed sample list **

FastQ file	Barcodes found in lane	Total # of reads per lane	Total number of good barcoded reads	Resulting # of Tags
C2YWBACXX_6_fastq.gz	96	239441926	188790120	15334349
C3008ACXX_8_fastq.gz	96	249840446	220293439	19640180

Number of Reads per Individual and Failed Samples

Failed samples (non-blank) are defined as those with less than 10% of the mean reads per sample coming from the lane on which they were sequenced. Blanks fail if they are greater than 10%.

Total number of samples: 192

Total number of failures: 10 (5.21%)

Flowcell	Lane	Barcode	ID	Good barcoded reads	Plate	Row	Col	status
C2YWBACXX	6	AACGCACATT	14	2601797	2_Hawaiian	B	2	pass
C2YWBACXX	6	AACGTGCCT	69	937658	2_Hawaiian	C	8	pass
C2YWBACXX	6	AACTGG	43	4691342	2_Hawaiian	E	5	pass
C2YWBACXX	6	AAGACGCT	99_18	4325392	2_Hawaiian	D	11	pass
C2YWBACXX	6	AATAACCAA	64	631179	2_Hawaiian	F	7	pass
C2YWBACXX	6	AATGAACGA	65	296722	2_Hawaiian	G	7	pass
C2YWBACXX	6	AATTAG	33	931944	2_Hawaiian	C	4	pass
C2YWBACXX	6	ACAACCAACT	207	2452451	2_Hawaiian	G	10	pass
C2YWBACXX	6	ACAACCT	35	1566572	2_Hawaiian	E	4	pass
C2YWBACXX	6	ACAGT	176	7314566	2_Hawaiian	H	9	pass
C2YWBACXX	6	ACCAGGA	47	1737573	2_Hawaiian	A	6	pass
C2YWBACXX	6	ACGCGCG	59	3251485	2_Hawaiian	D	7	pass
C2YWBACXX	6	ACGGTACT	73	4361625	2_Hawaiian	F	8	pass
C2YWBACXX	6	ACTGCGAT	25	551776	2_Hawaiian	D	3	pass
C2YWBACXX	6	ACTGCT	36	4279391	2_Hawaiian	F	4	pass
C2YWBACXX	6	ATAGAT	9	931759	2_Hawaiian	F	1	pass
C2YWBACXX	6	ATATAA	27	1039977	2_Hawaiian	F	3	pass
C2YWBACXX	6	ATATCGCCA	52	664986	2_Hawaiian	F	6	pass
C2YWBACXX	6	ATCCG	99_26	1730270	2_Hawaiian	G	11	pass
C2YWBACXX	6	ATCTGT	99_10	4276638	2_Hawaiian	C	11	pass

C2YWBACXX	6	ATGAGCAA		44	4061956	2_Hawaiian	F	5	pass
C2YWBACXX	6	ATGGCAA		67	1500470	2_Hawaiian	A	8	pass
C2YWBACXX	6	ATTAT		23	1242737	2_Hawaiian	B	3	pass
C2YWBACXX	6	CAACCACACA	D		9338893	2_Hawaiian	E	12	pass
C2YWBACXX	6	CAAGT		75	885740	2_Hawaiian	H	8	pass
C2YWBACXX	6	CACCA		32	414659	2_Hawaiian	B	4	pass
C2YWBACXX	6	CAGAGGT		56	1478102	2_Hawaiian	B	7	pass
C2YWBACXX	6	CAGATA		5	632776	2_Hawaiian	B	1	pass
C2YWBACXX	6	CAGTGCCATT		203	75100	2_Hawaiian	F	10	FAIL
C2YWBACXX	6	CATAT		194	1090833	2_Hawaiian	C	10	pass
C2YWBACXX	6	CATCTGCCG		212	540028	2_Hawaiian	A	11	pass
C2YWBACXX	6	CCACTCA		48	2552798	2_Hawaiian	B	6	pass
C2YWBACXX	6	CCGAACA		11	454485	2_Hawaiian	G	1	pass
C2YWBACXX	6	CCTCG		70	1647503	2_Hawaiian	D	8	pass
C2YWBACXX	6	CCTTGCCATT		17	8890	2_Hawaiian	D	2	FAIL
C2YWBACXX	6	CGCAACCAGT		86	1076191	2_Hawaiian	F	9	pass
C2YWBACXX	6	CGCACCAATT		41	2174628	2_Hawaiian	C	5	pass
C2YWBACXX	6	CGTCGCCACT		66	1331726	2_Hawaiian	H	7	pass
C2YWBACXX	6	CGTGGACAGT		37	1063621	2_Hawaiian	G	4	pass
C2YWBACXX	6	CGTGTCA	C		1688760	2_Hawaiian	D	12	pass
C2YWBACXX	6	CGTTCA	99_4		859488	2_Hawaiian	H	12	pass
C2YWBACXX	6	CTAAGCA		22	1882434	2_Hawaiian	A	3	pass
C2YWBACXX	6	CTCAT		72	4160727	2_Hawaiian	E	8	pass
C2YWBACXX	6	CTCGCGG		42	440546	2_Hawaiian	D	5	pass
C2YWBACXX	6	CTCGTCG		30	4115303	2_Hawaiian	H	3	pass
C2YWBACXX	6	CTCTA		53	1752447	2_Hawaiian	G	6	pass
C2YWBACXX	6	CTCTCGCAT		55	153007	2_Hawaiian	A	7	FAIL
C2YWBACXX	6	CTTAG	99_33		121968	2_Hawaiian	H	11	FAIL
C2YWBACXX	6	CTTGA		45	964169	2_Hawaiian	G	5	pass
C2YWBACXX	6	GAAGCA		68	604927	2_Hawaiian	B	8	pass
C2YWBACXX	6	GAAGTG		6	2181145	2_Hawaiian	C	1	pass
C2YWBACXX	6	GAATGCAATA	99_19		286893	2_Hawaiian	E	11	pass
C2YWBACXX	6	GAGCGACAT		16	1151455	2_Hawaiian	C	2	pass
C2YWBACXX	6	GCAAGCCAT		40	1731717	2_Hawaiian	B	5	pass
C2YWBACXX	6	GCACGAT		81	1435227	2_Hawaiian	D	9	pass
C2YWBACXX	6	GCCAACAAGA	A		4523998	2_Hawaiian	B	12	pass
C2YWBACXX	6	GCCTACCT		31	54796	2_Hawaiian	A	4	FAIL
C2YWBACXX	6	GCGCCG		74	259409	2_Hawaiian	G	8	pass
C2YWBACXX	6	GCGCTCA		24	1878777	2_Hawaiian	C	3	pass
C2YWBACXX	6	GCGTACAAT		58	1070148	2_Hawaiian	C	7	pass
C2YWBACXX	6	GCGTCCT		46	1412726	2_Hawaiian	H	5	pass

C2YWBACXX	6	GCTCCGA	E	3288743	2_Hawaiian	F	12	pass
C2YWBACXX	6	GGAACGA	34	5889996	2_Hawaiian	D	4	pass
C2YWBACXX	6	GGAAGACAT	12	3204566	2_Hawaiian	H	1	pass
C2YWBACXX	6	GGACAG	220	1918653	2_Hawaiian	B	11	pass
C2YWBACXX	6	GGAGTCAAG	188	1065849	2_Hawaiian	A	10	pass
C2YWBACXX	6	GGATA	21	1406293	2_Hawaiian	H	2	pass
C2YWBACXX	6	GGCTTA	13	529900	2_Hawaiian	A	2	pass
C2YWBACXX	6	GGTATA	18	1585743	2_Hawaiian	E	2	pass
C2YWBACXX	6	GGTGCACATT	54	3966455	2_Hawaiian	H	6	pass
C2YWBACXX	6	GGTGT	20	1235110	2_Hawaiian	G	2	pass
C2YWBACXX	6	GTCGCCT	63	189487	2_Hawaiian	E	7	FAIL
C2YWBACXX	6	GTGACACAT	197	2405181	2_Hawaiian	D	10	pass
C2YWBACXX	6	TAGATGA	79	10547	2_Hawaiian	B	9	FAIL
C2YWBACXX	6	TAGCAG	99_25	1528361	2_Hawaiian	F	11	pass
C2YWBACXX	6	TAGCCAA	51	1568884	2_Hawaiian	E	6	pass
C2YWBACXX	6	TAGCGGAT	7	5415002	2_Hawaiian	D	1	pass
C2YWBACXX	6	TATCA	50	4987256	2_Hawaiian	D	6	pass
C2YWBACXX	6	TATGT	201	4558094	2_Hawaiian	E	10	pass
C2YWBACXX	6	TATTCGCAT	8	472189	2_Hawaiian	E	1	pass
C2YWBACXX	6	TCACGGAAG	49	1684518	2_Hawaiian	C	6	pass
C2YWBACXX	6	TCACTG	88	1147588	2_Hawaiian	G	9	pass
C2YWBACXX	6	TCAGAGAT	99_34	176477	2_Hawaiian	G	12	FAIL
C2YWBACXX	6	TCCGAG	78	718976	2_Hawaiian	A	9	pass
C2YWBACXX	6	TCTTGG	19	1388590	2_Hawaiian	F	2	pass
C2YWBACXX	6	TGAAT	192	4579454	2_Hawaiian	B	10	pass
C2YWBACXX	6	TGACGCCA	4	1709767	2_Hawaiian	A	1	pass
C2YWBACXX	6	TGCAGA	210	1035814	2_Hawaiian	H	10	pass
C2YWBACXX	6	TGCCGCAT	B	630673	2_Hawaiian	C	12	pass
C2YWBACXX	6	TGCTT	39	1263475	2_Hawaiian	A	5	pass
C2YWBACXX	6	TGGCAACAGA	29	1587432	2_Hawaiian	G	3	pass
C2YWBACXX	6	TGGCACAGA	38	5441507	2_Hawaiian	H	4	pass
C2YWBACXX	6	TGGCCAG	80	3402986	2_Hawaiian	C	9	pass
C2YWBACXX	6	TTATTACAT	BLANK	8199	2_Hawaiian	A	12	pass
C2YWBACXX	6	TTCGTT	26	2469561	2_Hawaiian	E	3	pass
C2YWBACXX	6	TTGCTG	84	3443951	2_Hawaiian	E	9	pass
C3008ACXX	8	AACGCACATT	25_58	1560671	1_1025_Cross	B	2	pass
C3008ACXX	8	AACGTGCCT	25_248	1459074	1_1025_Cross	C	8	pass
C3008ACXX	8	AACTGG	25_130	1270499	1_1025_Cross	E	5	pass
C3008ACXX	8	AAGACGCT	25_418	3618115	1_1025_Cross	D	11	pass
C3008ACXX	8	AATAACCAA	25_234	2746071	1_1025_Cross	F	7	pass
C3008ACXX	8	AATGAACGA	25_237	2606704	1_1025_Cross	G	7	pass

C3008ACXX	8	AATTAG	25_111	884537	1_1025_Cross	C	4	pass
C3008ACXX	8	ACAACCAACT	25_377	1755840	1_1025_Cross	G	10	pass
C3008ACXX	8	ACAACCT	25_114	939840	1_1025_Cross	E	4	pass
C3008ACXX	8	ACAGT	25_299	2854327	1_1025_Cross	H	9	pass
C3008ACXX	8	ACCAGGA	25_175	2935148	1_1025_Cross	A	6	pass
C3008ACXX	8	ACGCGCG	25_225	1099798	1_1025_Cross	D	7	pass
C3008ACXX	8	ACGGTACT	25_256	2061286	1_1025_Cross	F	8	pass
C3008ACXX	8	ACTGCGAT	25_87	6057661	1_1025_Cross	D	3	pass
C3008ACXX	8	ACTGCT	25_118	4737977	1_1025_Cross	F	4	pass
C3008ACXX	8	ATAGAT	25_40	1589262	1_1025_Cross	F	1	pass
C3008ACXX	8	ATATAA	25_91	1863756	1_1025_Cross	F	3	pass
C3008ACXX	8	ATATCGCCA	25_188	3481155	1_1025_Cross	F	6	pass
C3008ACXX	8	ATCCG	25_482	4159212	1_1025_Cross	G	11	pass
C3008ACXX	8	ATCTGT	25_399	1252972	1_1025_Cross	C	11	pass
C3008ACXX	8	ATGAGCAA	25_131	2093639	1_1025_Cross	F	5	pass
C3008ACXX	8	ATGGCAA	25_240	1061233	1_1025_Cross	A	8	pass
C3008ACXX	8	ATTAT	25_82	4091699	1_1025_Cross	B	3	pass
C3008ACXX	8	CAACCACACA	25_511	47899	1_1025_Cross	E	12	FAIL
C3008ACXX	8	CAAGT	25_274	289187	1_1025_Cross	H	8	pass
C3008ACXX	8	CACCA	25_109	1394245	1_1025_Cross	B	4	pass
C3008ACXX	8	CAGAGGT	25_220	1095123	1_1025_Cross	B	7	pass
C3008ACXX	8	CAGATA	25_9	2065336	1_1025_Cross	B	1	pass
C3008ACXX	8	CAGTGCCATT	25_355	3895673	1_1025_Cross	F	10	pass
C3008ACXX	8	CATAT	25_327	3227526	1_1025_Cross	C	10	pass
C3008ACXX	8	CATCTGCCG	25_398	2142253	1_1025_Cross	A	11	pass
C3008ACXX	8	CACTCA	25_180	2592591	1_1025_Cross	B	6	pass
C3008ACXX	8	CCGAACA	25_44	4486162	1_1025_Cross	G	1	pass
C3008ACXX	8	CCTCG	25_250	963576	1_1025_Cross	D	8	pass
C3008ACXX	8	CCTTGCCATT	25_64	1470677	1_1025_Cross	D	2	pass
C3008ACXX	8	CGCAACCAGT	25_291	3516949	1_1025_Cross	F	9	pass
C3008ACXX	8	CGCACCAATT	25_125	1411170	1_1025_Cross	C	5	pass
C3008ACXX	8	CGTCGCCACT	25_239	2864680	1_1025_Cross	H	7	pass
C3008ACXX	8	CGTGGACAGT	25_119	560595	1_1025_Cross	G	4	pass
C3008ACXX	8	CGTGTCA	25_510	3363877	1_1025_Cross	D	12	pass
C3008ACXX	8	CGTTCA	BLANK	6785	1_1025_Cross	H	12	pass
C3008ACXX	8	CTAAGCA	25_80	2244619	1_1025_Cross	A	3	pass
C3008ACXX	8	CTCAT	25_255	2258675	1_1025_Cross	E	8	pass
C3008ACXX	8	CTCGCGG	25_129	3828040	1_1025_Cross	D	5	pass
C3008ACXX	8	CTCGTCG	25_99	497351	1_1025_Cross	H	3	pass
C3008ACXX	8	CTCTA	25_204	3982756	1_1025_Cross	G	6	pass
C3008ACXX	8	CTCTCGCAT	25_215	423403	1_1025_Cross	A	7	pass

C3008ACXX	8	CTTAG	25_492	3552694	1_1025_Cross	H	11	pass
C3008ACXX	8	CTTGA	25_168	4096187	1_1025_Cross	G	5	pass
C3008ACXX	8	GAAGCA	25_242	991232	1_1025_Cross	B	8	pass
C3008ACXX	8	GAAGTG	25_19	2931794	1_1025_Cross	C	1	pass
C3008ACXX	8	GAATGCAATA	25_473	2568210	1_1025_Cross	E	11	pass
C3008ACXX	8	GAGCGACAT	25_60	5227772	1_1025_Cross	C	2	pass
C3008ACXX	8	GCAAGCCAT	25_124	1161695	1_1025_Cross	B	5	pass
C3008ACXX	8	GCACGAT	25_287	3797288	1_1025_Cross	D	9	pass
C3008ACXX	8	GCCAACAAGA	25_507	2755651	1_1025_Cross	B	12	pass
C3008ACXX	8	GCCTACCT	25_100	1676700	1_1025_Cross	A	4	pass
C3008ACXX	8	GCGCCG	25_269	705069	1_1025_Cross	G	8	pass
C3008ACXX	8	GCGCTCA	25_83	2962312	1_1025_Cross	C	3	pass
C3008ACXX	8	GCGTACAAT	25_224	291403	1_1025_Cross	C	7	pass
C3008ACXX	8	GCGTCCT	25_174	5439162	1_1025_Cross	H	5	pass
C3008ACXX	8	GCTCCGA	P230	1584431	1_1025_Cross	F	12	pass
C3008ACXX	8	GGAACGA	25_113	1546152	1_1025_Cross	D	4	pass
C3008ACXX	8	GGAAGACAT	25_51	4627062	1_1025_Cross	H	1	pass
C3008ACXX	8	GGACAG	25_399	675265	1_1025_Cross	B	11	pass
C3008ACXX	8	GGAGTCAAG	25_312	1242158	1_1025_Cross	A	10	pass
C3008ACXX	8	GGATA	25_79	4442929	1_1025_Cross	H	2	pass
C3008ACXX	8	GGCTTA	25_56	1262982	1_1025_Cross	A	2	pass
C3008ACXX	8	GGTATA	25_71	4546323	1_1025_Cross	E	2	pass
C3008ACXX	8	GGTGCACATT	25_207	6561	1_1025_Cross	H	6	FAIL
C3008ACXX	8	GGTGT	25_74	899916	1_1025_Cross	G	2	pass
C3008ACXX	8	GTCGCCT	25_229	680204	1_1025_Cross	E	7	pass
C3008ACXX	8	GTGACACAT	25_332	2980692	1_1025_Cross	D	10	pass
C3008ACXX	8	TAGATGA	25_281	1417434	1_1025_Cross	B	9	pass
C3008ACXX	8	TAGCAG	25_480	2622634	1_1025_Cross	F	11	pass
C3008ACXX	8	TAGCCAA	25_187	1084029	1_1025_Cross	E	6	pass
C3008ACXX	8	TAGCGGAT	25_35	1836111	1_1025_Cross	D	1	pass
C3008ACXX	8	TATCA	25_186	1143036	1_1025_Cross	D	6	pass
C3008ACXX	8	TATGT	25_350	4025633	1_1025_Cross	E	10	pass
C3008ACXX	8	TATTCGCAT	25_37	2177273	1_1025_Cross	E	1	pass
C3008ACXX	8	TCACGGAAG	25_181	2244189	1_1025_Cross	C	6	pass
C3008ACXX	8	TCACTG	25_297	1324216	1_1025_Cross	G	9	pass
C3008ACXX	8	TCAGAGAT	P255	2140092	1_1025_Cross	G	12	pass
C3008ACXX	8	TCCGAG	25_278	5166141	1_1025_Cross	A	9	pass
C3008ACXX	8	TCTTGG	25_72	1491907	1_1025_Cross	F	2	pass
C3008ACXX	8	TGAAT	25_317	1040632	1_1025_Cross	B	10	pass
C3008ACXX	8	TGACGCCA	25_2	532667	1_1025_Cross	A	1	pass
C3008ACXX	8	TGCAGA	25_387	4049063	1_1025_Cross	H	10	pass

C3008ACXX	8	TGCCGCAT	25_509	3135608	1_1025_Cross	C	12	pass
C3008ACXX	8	TGCTT	25_122	838679	1_1025_Cross	A	5	pass
C3008ACXX	8	TGGCAACAGA	25_96	1324835	1_1025_Cross	G	3	pass
C3008ACXX	8	TGGCACAGA	25_120	6820142	1_1025_Cross	H	4	pass
C3008ACXX	8	TGGCCAG	25_283	991905	1_1025_Cross	C	9	pass
C3008ACXX	8	TTATTACAT	25_502	913324	1_1025_Cross	A	12	pass
C3008ACXX	8	TTCGTT	25_89	5062994	1_1025_Cross	E	3	pass
C3008ACXX	8	TTGCTG	25_288	1449734	1_1025_Cross	E	9	pass

Summary of Resulting SNPs

- HapMap SNPs (unfiltered) : 7362
- HapMap SNPs (filtered) : 1368
- VCF SNPs: 19395

VCFtools version [v0.1.10] was used to calculate Depth and Missingness from the file all.mergedSNPs.vcf.gz

	mean	median	standard_deviation
individual depth	6.555	5.944	3.677
site depth	6.555	0.337	20.948
individual missingness	0.765	0.759	0.082
site missingness	0.765	0.916	0.293

Multi Dimension Scaling (MDS) of genome-wide SNPs

VCFTools version [v0.1.10] and Plink version [v1.07] were used to generate an MDS plot

- Genotypes were filtered to those with genotype quality 98 or higher (high confidence SNP calls) with VCFTools
- Biallelic SNPs were converted to PLINK format
- The remaining individuals/sites were filtered on missingness and allele frequency (with the same parameters as those used for the GBSHapMapFiltersPlugin
- Individuals removed from the analysis because of too much missing data:
 - 11:C2YWBACXX:6:250249534
 - 13:C2YWBACXX:6:250249536
 - 17:C2YWBACXX:6:250249539
 - 188:C2YWBACXX:6:250249600
 - 194:C2YWBACXX:6:250249602
 - 203:C2YWBACXX:6:250249605
 - 212:C2YWBACXX:6:250249608
 - 25:C2YWBACXX:6:250249547
 - 25_111:C3008ACXX:8:250249427
 - 25_186:C3008ACXX:8:250249444
 - 25_2:C3008ACXX:8:250249401
 - 25_207:C3008ACXX:8:250249448
 - 25_215:C3008ACXX:8:250249449
 - 25_224:C3008ACXX:8:250249451
 - 25_225:C3008ACXX:8:250249452
 - 25_229:C3008ACXX:8:250249453
 - 25_250:C3008ACXX:8:250249460
 - 25_269:C3008ACXX:8:250249463
 - 25_274:C3008ACXX:8:250249464
 - 25_502:C3008ACXX:8:250249489
 - 25_511:C3008ACXX:8:250249493
 - 25_99:C3008ACXX:8:250249424
 - 27:C2YWBACXX:6:250249549
 - 31:C2YWBACXX:6:250249552
 - 32:C2YWBACXX:6:250249553
 - 33:C2YWBACXX:6:250249554
 - 39:C2YWBACXX:6:250249560
 - 42:C2YWBACXX:6:250249563
 - 52:C2YWBACXX:6:250249573
 - 55:C2YWBACXX:6:250249576
 - 56:C2YWBACXX:6:250249577
 - 58:C2YWBACXX:6:250249578

- 63:C2YWBACXX:6:250249580
 - 64:C2YWBACXX:6:250249581
 - 65:C2YWBACXX:6:250249582
 - 67:C2YWBACXX:6:250249584
 - 68:C2YWBACXX:6:250249585
 - 69:C2YWBACXX:6:250249586
 - 70:C2YWBACXX:6:250249587
 - 74:C2YWBACXX:6:250249590
 - 75:C2YWBACXX:6:250249591
 - 78:C2YWBACXX:6:250249592
 - 79:C2YWBACXX:6:250249593
 - 8:C2YWBACXX:6:250249532
 - 86:C2YWBACXX:6:250249597
 - 88:C2YWBACXX:6:250249598
 - 9:C2YWBACXX:6:250249533
 - 99_19:C2YWBACXX:6:250249612
 - 99_33:C2YWBACXX:6:250249615
 - 99_34:C2YWBACXX:6:250249622
 - 99_4:C2YWBACXX:6:250249623
 - BLANK:MERGE
- Number of remaining sites for analysis: [1108] filtered, biallelic SNPs ****Warning**** this analysis typically requires whole-genome level data to give accurate results

Genotyping by Sequencing Resources

GBS Overview

<http://www.igd.cornell.edu/index.cfm/page/GBS.htm>

<http://www.maizegenetics.net/gbs-overview>

GBS Frequently Asked Questions and IGD policies

<http://www.igd.cornell.edu/index.cfm/page/GBS/gbsfaq.htm>

<http://www.igd.cornell.edu/index.cfm/page/GBS/GBSpoliciesbrief.htm>

GBS Bioinformatics

<http://www.maizegenetics.net/gbs-bioinformatics>

Training

<http://www.igd.cornell.edu/index.cfm/page/Education/workshops/GBSworkshop.htm>

TASSEL

General Information

<http://sourceforge.net/projects/tassel/>

http://www.maizegenetics.net/index.php?option=com_content&task=view&id=89&Itemid=119

Tassel 3.0 GBS pipeline documentation

<http://www.maizegenetics.net/tassel/docs/TasselPipelineGBS.pdf>

Tassel 3.0/4.0 pipeline documentation

<http://www.maizegenetics.net/tassel/docs/TasselPipelineCLI.pdf>

Tassel Users Group

<https://groups.google.com/forum/?fromgroups#!forum/tassel>

Bibliography:

Elshire RJ, Glaubitz JC, Sun Q, Poland JA, Kawamoto K, et al. (2011) A Robust, Simple Genotyping-by-Sequencing (GBS) Approach for High Diversity Species. PLoS ONE 6(5): e19379. doi:10.1371/journal.pone.0019379

Bradbury, P. J., Z. Zhang, D. E. Kroon, T. M. Casstevens, Y. Ramdoss, and E. S. Buckler. "TASSEL: Software for Association Mapping of Complex Traits in Diverse Samples." Bioinformatics 23, no. 19 (June 22, 2007): 2633–2635.

Glaubitz JC, Casstevens TM, Lu F, Harriman J, Elshire RJ, Sun Q and Buckler ES. (2013) TASSEL-GBS: A High Capacity Genotyping by Sequencing Analysis Pipeline. PLoS ONE (in press).

Lu, F, Lipka, AE, Glaubitz J, Elshire R, Cherney JH, Casler MD, Buckler ES and Costich DE. "Switchgrass Genomic Diversity, Ploidy, and Evolution: Novel Insights from a Network-Based SNP Discovery Protocol". PLoS Genet 9(1):e1003215. Doi:10.1371/journal.pgen.1003215

J. Catchen, A. Amores, P. Hohenlohe, W. Cresko, and J. Postlethwait. *Stacks: building and genotyping loci de novo from short-read sequences*. **G3: Genes, Genomes, Genetics**, 1:171-182, 2011

Petr Danecek, Adam Auton, Goncalo Abecasis, Cornelis A. Albers, Eric Banks, Mark A. DePristo, Robert Handsaker, Gerton Lunter, Gabor Marth, Stephen T. Sherry, Gilean McVean, Richard Durbin and 1000 Genomes Project Analysis Group. The Variant Call Format and VCFtools. Bioinformatics, 2011

Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MAR, Bender D, Maller J, Sklar P, de Bakker PIW, Daly MJ & Sham PC (2007) PLINK: a toolset for whole-genome association and population-based linkage analysis. American Journal of Human Genetics, 81, URL:<http://pngy.mgh.harvard.edu/purcell/plink>

R Development Core Team (2008). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL <http://www.R-project.org>.